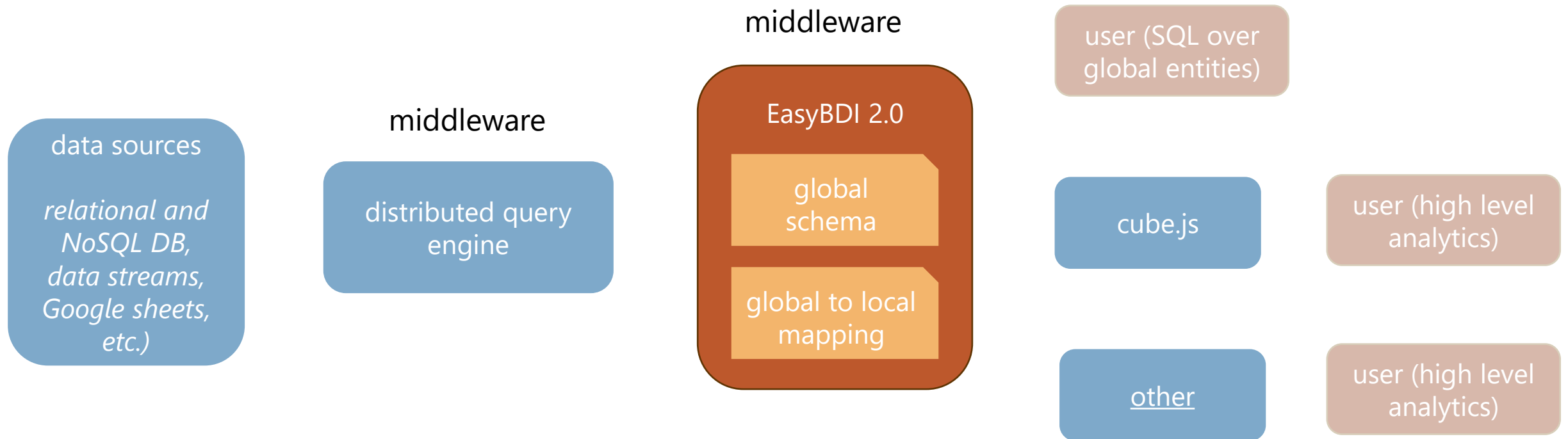


Homework assignments



HW02.1: EasyBDI 2.0: Big Data Integration



HW02.1: EasyBDI 2.0: Big Data Integration

Postgres: customers_pg table

CustomerID	Name	Country
3	Charlie	Canada
4	Diana	Canada
5	Eve	UK

MongoDB: customers collection

```
[  
  { _id: 3, Email: "charlie@x.com", Age: 28 },  
  { _id: 4, Email: "diana@x.com", Age: 35 },  
  { _id: 5, Email: "eve@x.com", Age: 22 }  
]
```

CSV or Excel file

CustomerID	Name	Email	Country	Age
1	Alice	alice@x.com	USA	25
2	Bob	bob@x.com	USA	30

Global table or view : allCustomers

CustomerID	Name	Email	Country	Age
1	Alice	alice@x.com	USA	25
2	Bob	bob@x.com	USA	30
3	Charlie	charlie@x.com	Canada	28
4	Diana	diana@x.com	Canada	35
5	Eve	eve@x.com	UK	22

Assuming the three datasets shown on the left (white background) are stored in separate databases or files, the goal is to provide a unified, global view of these data as represented in the gray table at the bottom left. This allows developers to run queries such as `'SELECT * FROM allCustomers'` and retrieve the complete dataset without needing to understand the underlying data structures or system-specific details.

This is a typical example of horizontal and vertical data partitioning. For additional details refer to <https://openproceedings.org/2021/conf/edbt/p190.pdf>.

HW02.2 E-Procurement

- Clients: publish tenders
 - Portal BASE: <https://www.base.gov.pt/base4>
 - VORTAL GOV: <https://www.vortal.biz/pt-pt/vortal-gov/>
 - AcinGov: <https://www.acingov.pt/acingovprod/2/index.php/>
 - AnoGov: <https://anogov.com/home/en/home-english/>
 - ComprasPT: <https://compraspt.com/home/>
 - EU Tenders: <https://ted.europa.eu/en/>
 - Open Tender France: <https://opentender.eu/fr/search/tender>
 - AMT: <https://www.amf-france.org/en/tenders>
 - PDF files with additional information
- Suppliers: submit bids
 - What are the tenders of interest?
 - Search a curated database using dashboards or natural language
 - get a daily list of tenders of interest

